

Spring Core and Maven

Exercise 1: Configuring a Basic Spring Application

FileEditSelectionViewGoRun...spring_coreSign in

EXPLORER

SPRING CORELibrary_managementsrcmainjavacomlibraryrepositoryBookRepository.javaBookService.javaMainApp.javaapplicationContext.xmltargetpom.xml

BookRepository.java

Library_management > src > main > java > com > library > repository > BookRepository.java1package com.library.repository;23public class BookRepository {45public void init() {6System.out.println("[BookRepository] Bean initialized - ready to connect to data source.");7}89public String findBookById(int id) {10// In a real app: return bookJpaRepository.findById(id)11System.out.println("[BookRepository] Fetching book with ID: " + id);12return "Book #" + id + " → 'Clean Code' by Robert C. Martin";13}1415public void saveBook(String title) {16System.out.println("[BookRepository] Saving book: " + title);17}18}

BookService.java

Library_management > src > main > java > com > library > service > BookService.javaimport com.library.repository.BookRepository;12public class BookService {34private BookRepository bookRepository;56public void setBookRepository(BookRepository bookRepository) {7this.bookRepository = bookRepository;8System.out.println("[BookService] BookRepository injected successfully.");9}1011public BookRepository getBookRepository() {12return bookRepository;13}1415public void init() {16System.out.println("[BookService] Bean initialized - all dependencies ready.");17}1819public void findBook(int id) {20System.out.println("[BookService] Request to find book ID: " + id);21// ❌ If bookRepository was never injected (setter missing / XML misconfigured),22// this line throws a NullPointerException23String result = bookRepository.findBookById(id);24System.out.println("[BookService] Result: " + result);25}2627public void addBook(String title) {

LEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Library_management > src > main > java > com > library > MainApp.java

package com.library;

import com.library.service.BookService;

import org.springframework.context.ApplicationContext;

import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {

public static void main(String[] args) {

System.out.println("=== Starting Spring Application Context ===\n");

ApplicationContext context =

new ClassPathXmlApplicationContext("applicationContext.xml");

System.out.println("\n=== Context Loaded - Fetching Beans ===\n");

BookService bookService = (BookService) context.getBean("bookService");

System.out.println("\n=== Calling Business Methods ===\n");

bookService.findBook(101);

System.out.println();

bookService.addBook("The Pragmatic Programmer");

System.out.println("\n=== Closing Application Context ===");

((ClassPathXmlApplicationContext) context).close();

}

```
BookRepository.java pom.xml BookService.java MainApp.java applica
Library_management > pom.xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
    http://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.library</groupId>
  <artifactId>LibraryManagement</artifactId>
  <version>1.0-SNAPSHOT</version>
  <packaging>jar</packaging>

  <properties>

    <maven.compiler.source>11</maven.compiler.source>
    <maven.compiler.target>11</maven.compiler.target>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-core</artifactId>
      <version>5.3.30</version>
    </dependency>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-context</artifactId>
      <version>5.3.30</version>
    </dependency>

    <dependency>
      <groupId>org.springframework</groupId>
      <artifactId>spring-beans</artifactId>
      <version>5.3.30</version>
    </dependency>

  </dependencies>
</project>

[INFO] Skip non-existing resource directory C:\Users\SARO\Downloads\Spring_core\Library_management\src\main
[INFO] --- compiler:3.15.0:compile (default-compile) @ LibraryManagement ---
[INFO] Recompiling the module because of changed source code.
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent
[INFO] Compiling 3 source files with javac [debug target 11] to target\classes
[WARNING] location of system modules is not set in conjunction with -source 11
    not setting the location of system modules may lead to class files that cannot run on JDK 11
    --release 11 is recommended instead of -source 11 -target 11 because it sets the location of system
    modules
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 2.608 s
[INFO] Finished at: 2026-07-01T05:19:14+05:30
[INFO] -----
PS C:\Users\SARO\Downloads\Spring_core\Library_management> |
```

Exercise 2: Implementing Dependency Injection

```
Library_management > src > main > resources > applicationContext.xml
1 <?xml version="1.0" encoding="UTF-8"?>
2
3 <beans xmlns="http://www.springframework.org/schema/beans"
4   xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
5   xsi:schemaLocation="
6     http://www.springframework.org/schema/beans
7     http://www.springframework.org/schema/beans/spring-beans.xsd">
8
9
10
11   <bean id="bookRepository"
12     class="com.library.repository.BookRepository"
13     init-method="init">
14
15
16
17   <bean id="bookService"
18     class="com.library.service.BookService"
19     init-method="init">
20
21
22     <property name="bookRepository" ref="bookRepository"/>
23
24   </bean>
25
26 </beans>
```

```
Library_management > src > main > java > com > library > MainApp.java
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class MainApp {

    public static void main(String[] args) {

        System.out.println("=====");
        System.out.println(" Spring DI - Library Management App ");
        System.out.println("=====\\n");

        ApplicationContext context =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        System.out.println("\\n--- Fetching BookService Bean ---\\n");

        BookService bookService = (BookService) context.getBean("bookService");

        if (bookService.getBookRepository() != null) {
            System.out.println(" DI SUCCESS: BookRepository is injected into BookService");
        } else {
            System.out.println(" DI FAILED: BookRepository is NULL");
        }

        System.out.println("\\n--- Testing Business Methods ---");

        bookService.findBook(101);
    }
}
```

```
management> src> main> java> com> library> service> BookService.java
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;

    public void setBookRepository(BookRepository bookRepository) {
        this.bookRepository = bookRepository;
        System.out.println("[BookService] BookRepository injected via Setter DI.");
    }

    public BookRepository getBookRepository() {
        return bookRepository;
    }

    public void init() {
        System.out.println("[BookService] All dependencies injected. Service ready.");
    }

    public void findBook(int id) {
        System.out.println("\n[BookService] Finding book with ID: " + id);
        String result = bookRepository.findBookById(id);
        System.out.println("[BookService] Found: " + result);
    }

    public void addBook(String title) {
```

```
[BookRepository] Fetching book with ID: 101
[BookService] Found: Book #101 ? 'Clean Code' by Robert C. Martin

[BookService] Adding book: 'The Pragmatic Programmer'
[BookRepository] Saving book: The Pragmatic Programmer
[BookService] Book added successfully.

[BookService] Finding book with ID: 202
[BookRepository] Fetching book with ID: 202
[BookService] Found: Book #202 ? 'Clean Code' by Robert C. Martin

=====
Closing Spring Context
=====
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.711 s
[INFO] Finished at: 2026-07-01T05:42:25+05:30
[INFO] -----
PS C:\Users\SABO\Downloads\Spring-core\Library-management>
```

Exercise 3: Creating and Configuring a Maven Project

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <project xmlns="http://maven.apache.org/POM/4.0.0"
3     xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4     xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
5         http://maven.apache.org/xsd/maven-4.0.0.xsd">
6
7     <modelVersion>4.0.0</modelVersion>
8
9
10    <groupId>com.library</groupId>
11    <artifactId>LibraryManagement</artifactId>
12    <version>1.0-SNAPSHOT</version>
13    <packaging>jar</packaging>
14
15
16    <name>LibraryManagement</name>
17    <description>Library Management Application using Spring Framework</description>
18
19    <properties>
20
21        <maven.compiler.source>1.8</maven.compiler.source>
22        <maven.compiler.target>1.8</maven.compiler.target>
23
24
25        <spring.version>5.3.30</spring.version>
26
27
28        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
29    </properties>
30
31
32    <dependencies>
33
34
35        <dependency>
36            <groupId>org.springframework</groupId>
37            <artifactId>spring-context</artifactId>
38            <version>${spring.version}</version>
39        </dependency>
40
41
42        <dependency>
43            <groupId>org.springframework</groupId>
44            <artifactId>spring-aop</artifactId>
45            <version>${spring.version}</version>
46        </dependency>
47
48
49        <dependency>
50            <groupId>org.springframework</groupId>
51            <artifactId>spring-webmvc</artifactId>
52            <version>${spring.version}</version>
53        </dependency>
54
55
56        <dependency>
57            <groupId>org.springframework</groupId>
58            <artifactId>spring-core</artifactId>
59            <version>${spring.version}</version>
60        </dependency>
61
62
63        <dependency>
```

```
Library_management > pom.xml
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
32  <dependencies>
61
62      <dependency>
63          <groupId>org.springframework</groupId>
64          <artifactId>spring-beans</artifactId>
65          <version>${spring.version}</version>
66      </dependency>
67
68      <dependency>
69          <groupId>javax.servlet</groupId>
70          <artifactId>javax.servlet-api</artifactId>
71          <version>4.0.1</version>
72          <scope>provided</scope>
73      </dependency>
74
75      <dependency>
76          <groupId>junit</groupId>
77          <artifactId>junit</artifactId>
78          <version>4.13.2</version>
79          <scope>test</scope>
80      </dependency>
81
82  </dependencies>
83
84  <build>
85      <plugins>
86
87          <plugin>
88              <groupId>org.apache.maven.plugins</groupId>
89              <artifactId>maven-compiler-plugin</artifactId>
90              <version>3.11.0</version>
91              <configuration>
92                  <source>1.8</source>
93                  <target>1.8</target>
94                  <encoding>UTF-8</encoding>
95              </configuration>
96          </plugin>
97
98          <plugin>
99              <groupId>org.codehaus.mojo</groupId>
100              <artifactId>exec-maven-plugin</artifactId>
101              <version>3.1.0</version>
102              <configuration>
103                  <mainClass>com.library.MainApp</mainClass>
104              </configuration>
105          </plugin>
106      </plugins>
107  </build>
108
109  </project>
```

```
s)
[INFO] com.library:LibraryManagement:jar:1.0-SNAPSHOT
[INFO] +- org.springframework:spring-context:jar:5.3.30:compile
[INFO] |   \- org.springframework:spring-expression:jar:5.3.30:compile
[INFO] +- org.springframework:spring-aop:jar:5.3.30:compile
[INFO] +- org.springframework:spring-webmvc:jar:5.3.30:compile
[INFO] |   \- org.springframework:spring-web:jar:5.3.30:compile
[INFO] +- org.springframework:spring-core:jar:5.3.30:compile
[INFO] |   \- org.springframework:spring-jcl:jar:5.3.30:compile
[INFO] +- org.springframework:spring-beans:jar:5.3.30:compile
[INFO] +- javax.servlet:javax.servlet-api:jar:4.0.1:provided
[INFO] \- junit:junit:jar:4.13.2:test
[INFO]     \- org.hamcrest:hamcrest-core:jar:1.3:test
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 16.302 s
[INFO] Finished at: 2026-07-01T05:48:20+05:30
[INFO] -----
PS C:\Users\SARO\Downloads\Spring_core\Library_management> 
```